

## Multiple Choice (10 marks)

1.  $^{23}\text{Na}$  has an NMR frequency of 198.52 MHz in a 750 MHz NMR spectrometer. What is its magnetogyric ratio?
  - (a)  $101.1 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$
  - (b)  $26.75 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$
  - (c)  $7.081 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$
  - (d)  $0.9377 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$
2. Which of the following statements most accurately explains why the  $T_1$  of a nucleus is sometimes longer than its  $T_2$ ?
  - (a)  $T_1$ , unlike  $T_2$ , is sensitive to very low-frequency molecular motions.
  - (b)  $T_2$ , unlike  $T_1$ , is sensitive to very low-frequency molecular motions.
  - (c)  $T_1$ , unlike  $T_2$ , is sensitive to molecular motions at the Larmor frequency.
  - (d)  $T_2$ , unlike  $T_1$ , is sensitive to molecular motions at the Larmor frequency.
3. The normal modes of a carbon dioxide molecule that are infrared-active include which of the following? I. Bending II. Symmetric stretching III. Asymmetric stretching
  - (a) I only
  - (b) II only
  - (c) III only
  - (d) I and III only
4. Cobalt-60 is used in the radiation therapy of cancer and can be produced by bombardment of cobalt-59 with which of the following?
  - (a) Neutrons
  - (b) Alpha particles
  - (c) Beta particles
  - (d) X-rays
5. Which of the following types of spectroscopy is a light-scattering technique?
  - (a) Nuclear magnetic resonance
  - (b) Infrared
  - (c) Raman
  - (d) Ultraviolet-visible

## True / False (10 marks)

*Answer True or False*

6. True or False: Both  $^{35}\text{Cl}$  and  $^{37}\text{Cl}$  possess electric quadrupole moments, which contribute to the singlet appearance of the  $^1\text{H}$  NMR spectrum of  $\text{CHCl}_3$ .
7. True or False: At standard temperature and pressure,  $\text{Br}_2$  is a colorless gas.
8. True or False: The  $^{55}\text{Mn}$  nuclide has three allowed energy levels for its nuclear states.
9. True or False: Gadolinium (Gd) has fully filled 4 f orbitals, making it an exception among the lanthanides.
10. True or False: The equilibrium polarization of  $^{13}\text{C}$  nuclei in a 20.0 T magnetic field at 300 K is  $1.71 \times 10^{-5}$ .

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the stability of the +1 and +3 oxidation states in group 13 elements, focusing on thallium. What factors contribute to the greater stability of the +1 state in this element?
12. In the reaction  $3 \text{Cl}^-(\text{aq}) + 4 \text{CrO}_4^{2-}(\text{aq}) + 23 \text{H}^+(\text{aq}) \rightarrow 3 \text{HClO}_2(\text{aq}) + 4 \text{Cr}^{3+}(\text{aq}) + 10 \text{H}_2\text{O}(\text{l})$ , analyze the role of  $\text{Cl}^-(\text{aq})$  and discuss how it functions as a reducing agent.

*End of Paper*